

Divide Mixed Numbers

Lesson 2-4

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$2\frac{1}{3}$ means 2 wholes and $\frac{1}{3}$ more — picture 2 full circles and $\frac{1}{3}$ of another

Mixed Number

A whole number plus a fraction, like $2\frac{1}{2}$.

$\frac{7}{3}$ is improper because $7 > 3$. It equals $2\frac{1}{3}$ as a mixed number.

Improper Fraction

A fraction where the top is bigger than or equal to the bottom, like $\frac{7}{4}$.

$2\frac{1}{3} \rightarrow$ multiply $2 \times 3 = 6$, add 1 $\rightarrow \frac{7}{3}$. Same value, different form.

Convert

To change a number to a new form but keep the same value.

$\frac{8}{6}$: GCF is 2 $\rightarrow \frac{8 \div 2}{6 \div 2} = \frac{4}{3} = 1\frac{1}{3}$

Simplify

To make a fraction smaller using the same parts, like $\frac{2}{4} = \frac{1}{2}$.

The reciprocal of $\frac{3}{4}$ is $\frac{4}{3}$. To divide by $\frac{3}{4}$, multiply by $\frac{4}{3}$.

Reciprocal

A fraction turned upside down. You use it in keep-change-flip.

Key Ideas & Notes

- Agent Chen is mapping a suspect's route.
- The total distance is $3\frac{1}{2}$ miles, and each segment between landmarks is $\frac{1}{4}$ mile.
- She needs to know how many segments are on the route to place surveillance cameras at every landmark.
- Convert each mixed number to an improper fraction, then use Keep, Change, Flip to divide.

Think About It

- What is the total distance of the route?
- How long is each segment?
- Will the number of segments be more or less than 3?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Agent Chen maps a $3\frac{1}{2}$ -mile route divided into $\frac{1}{4}$ -mile segments. Why can't she just divide $3\frac{1}{2}$ by $\frac{1}{4}$ while it is still a mixed number?

Level 1 support

Start here: A mixed number must first become an improper fraction so I can flip and multiply.

I cannot divide yet because $3\frac{1}{2}$ is a ____ and I need an ____.

No puedo dividir todavía porque $3\frac{1}{2}$ es un ____ y necesito una ____.

First I must convert $3\frac{1}{2}$ into ____.

Primero debo convertir $3\frac{1}{2}$ en ____.

Word bank: **mixed number** **improper fraction** **convert**

Listen for: Students explain that the mixed number must be converted to an improper fraction ($\frac{7}{2}$) before applying Keep, Change, Flip.

Level 2

Estimate the number of segments first. Is your estimate close to 14? Justify using the route length and segment size.

- I estimate about ____ segments because ____.
- Each mile holds ____ quarter-mile segments, so ____.

EXPLORE

Walk your partner through how you changed the mixed numbers before dividing.

Level 1 support

Start here: I converted each mixed number to an improper fraction, then used Keep, Change, Flip.

First I converted ____ **into the improper fraction** ____.

Primero convertí ____ en la fracción impropia ____.

Then I flipped the divisor and got the quotient ____.

Luego volteé el divisor y obtuve el cociente ____.

Word bank: **mixed number** **improper fraction** **reciprocal** **simplify**

Listen for: Students correctly convert (multiply whole by denominator, add numerator) before flipping the divisor and simplifying.

Level 2

A student converted $2\frac{1}{3}$ to $\frac{6}{3}$ instead of $\frac{7}{3}$. Critique exactly what step they skipped.

- The student forgot to ____.
- The correct improper fraction is ____ because ____.

CONNECT

When would dividing mixed numbers come up in a real project, like building or cooking?

Level 1 support

Start here: Whenever you split a mixed-number amount into fraction-size pieces.

I would divide mixed numbers when ____.

Dividiría números mixtos cuando ____.

For example, cutting ____ feet into ____-foot pieces gives ____.

Por ejemplo, cortar ____ pies en piezas de ____ pie da ____.

Word bank: **mixed number** **improper fraction** **convert**

Listen for: Students give a realistic project scenario and recognize the mixed-number amount must be converted first.

Level 2

Generalize the full set of steps for dividing any two mixed numbers, in order.

- My steps are: first ____, then ____, then ____.
- This always works because ____.

Guided Examples

Example 1

What is $1\frac{1}{2} \div \frac{3}{4}$?

Solution: $1\frac{1}{2} = \frac{3}{2}$. Then $\frac{3}{2} \div \frac{3}{4} = \frac{3}{2} \times \frac{4}{3} = \frac{12}{6} = 2$.

Answer: A. 2

Example 2

Convert $3\frac{2}{5}$ to an improper fraction.

Solution: $3 \times 5 = 15$, then $15 + 2 = 17$. So $3\frac{2}{5} = \frac{17}{5}$.

Answer: A. $\frac{17}{5}$

Example 3

What is $2\frac{1}{4} \div \frac{3}{4}$?

Solution: $2\frac{1}{4} = \frac{9}{4}$. Then $\frac{9}{4} \div \frac{3}{4} = \frac{9}{4} \times \frac{4}{3} = \frac{36}{12} = 3$.

Answer: A. 3

Write About the Math

The Writing Revolution

I can explain my steps using the words mixed number, improper fraction, convert, reciprocal, and simplify.

1. Kernel Sentence subject + verb

Model: Mixed Number is a whole number plus a fraction, like $2\frac{1}{2}$.

Número mixto es un número entero más una fracción, como $2\frac{1}{2}$.

Write a kernel sentence about mixed number. Use a subject and a verb.

Escribe una oración base sobre número mixto. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Mixed Number matters in math

Número mixto importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Mixed Number matters in math because ____.

Número mixto importa en matemáticas porque ____.

but
pero

Mixed Number matters in math, but ____.

Número mixto importa en matemáticas, pero ____.

so
entonces

Mixed Number matters in math, so ____.

Número mixto importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about mixed number.
Di un hecho verdadero sobre mixed number.

Mixed number ____.

Question *Pregunta*

Ask a question about mixed number.
Haz una pregunta sobre mixed number.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about mixed number.
Muestra entusiasmo sobre mixed number.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with mixed number.
Dile a un compañero qué hacer con mixed number.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

I cannot divide yet because $3\frac{1}{2}$ is a ____ and I need an ____.

No puedo dividir todavía porque $3\frac{1}{2}$ es un ____ y necesito una ____.

First I must convert $3\frac{1}{2}$ into ____.

Primero debo convertir $3\frac{1}{2}$ en ____.

First I converted ____ into the improper fraction ____.

Primero convertí ____ en la fracción impropia ____.

Try It

Solve on your own. Check the answer key when you are done.

1. What is $3\frac{1}{2} \div \frac{1}{4}$?

- A. 14
- B. $3\frac{1}{8}$
- C. $\frac{7}{8}$
- D. $\frac{13}{4}$

Show your work:

2. A recipe makes $2\frac{1}{4}$ batches of cookies. Each batch uses $\frac{3}{4}$ cup of butter. How many cups of butter are needed in total?

- A. 3 cups
- B. $1\frac{1}{2}$ cups
- C. 2 cups
- D. 6 cups

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A detective has a rope that is $5 \frac{1}{4}$ feet long. She needs to cut it into pieces that are each $\frac{3}{4}$ foot for tying evidence tags. How many pieces can she cut? Write the equation, show each step (convert, keep-change-flip, multiply, simplify), and verify your answer.

Sentence starter: First I convert $5 \frac{1}{4}$ to _____. Then I use KCF: _____ $\times$ _____ = _____. The answer is _____ pieces. I can verify because _____ $\times \frac{3}{4} = 5 \frac{1}{4}$.

Show your work:

Reflect — Exit Ticket

What is $2 \frac{2}{3} \div \frac{2}{3}$?

- A. 4
- B. $\frac{2}{3}$
- C. $\frac{8}{3}$
- D. $1 \frac{7}{9}$

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. $14 - 3\frac{1}{2} = 7\frac{1}{2}$. Then $7\frac{1}{2} \div \frac{1}{4} = 7\frac{1}{2} \times \frac{4}{1} = 28\frac{2}{2} = 14$.
2. **Try It 2:** A. 3 cups — *This is multiplication: $2\frac{1}{4} \times \frac{3}{4} = \frac{9}{4} \times \frac{3}{4} = \frac{27}{16}$. Wait — re-read: we need $2\frac{1}{4}$ batches at $\frac{3}{4}$ cup each, but the question for division practice should be: if you have $2\frac{1}{4}$ cups and each batch needs $\frac{3}{4}$ cup, how many batches? $2\frac{1}{4} \div \frac{3}{4} = \frac{9}{4} \times \frac{4}{3} = \frac{36}{12} = 3$.*
3. **Exit Ticket:** A. $4 - 2\frac{2}{3} = \frac{8}{3}$. Then $\frac{8}{3} \div \frac{2}{3} = \frac{8}{3} \times \frac{3}{2} = \frac{24}{6} = 4$.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Mixed Number is a whole number plus a fraction, like $2\frac{1}{2}$.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).